Identities used: binomial therom
$(x+a)^r = \sum_{b=0}^{r} \left[\binom{r}{b} x^b a^{r-b} \right]$
pascals identity
$\binom{a}{n} = \binom{a-1}{n} + \binom{a-1}{n-1}$

Reucrsions from Binomial therom

Non falling sums

$$(x+a)^r = \sum_{b=0}^{r} \left[\binom{r}{b} x^b a^{r-b} \right]$$

let $a = 1$

$$(x+1)^r = \sum_{b=0}^{r} \left[\binom{r}{b} x^b \right]$$

let $x = x' - 1$

$$((x'-1)+1)^r = x^r = \sum_{b=0}^{r} \left[\binom{r}{b} (x'-1)^b \right]$$

in general

$$((x-b)+a)^r = \sum_{b=0}^{r} \left[\binom{r}{b} (x-b)^b a^{r-b} \right]$$

if b=a

$$x^r = \sum_{b=0}^{r} \left[\binom{r}{b} (x-a)^b a^{r-b} \right]$$

we can subtitute the defintion of x^r into $(x-a)^b$ or a^{r-b}
$(x-a)^b$ is a good substitution as it the nested power term will shrink by a
after $\frac{x}{a}$ expansions the inner power term will be 0

$$x^r = \sum_{b=0}^{r}\left[\binom{r}{b}(x-a)^b a^{r-b}\right]$$

$$(x-a)^b = \sum_{c=0}^{b}\left[\binom{b}{c}((x-a)-a)^c a^{b-c}\right]$$

$$x^r = \sum_{b=0}^{r}\left[\binom{r}{b}\sum_{c=0}^{b}\left[\binom{b}{c}((x-a)-a)^c a^{b-c}\right]a^{r-b}\right]$$

every layer strips one a from inner power's base

$$x^r = \sum_{b=0}^{r}\left[\binom{r}{b}\sum_{c=0}^{b}\left[\binom{b}{c}((x-a)-a)^c a^{b-c}\right]a^{r-b}\right]$$

$$x^r = \overbrace{\sum_{b=0}^{r}\left[\binom{r}{b}\sum_{c=0}^{b}\left[\binom{b}{c}\sum_{d=0}^{c}\left[\binom{c}{d}\cdots\left[\sum_{n_{\frac{x}{a}}=0}^{n_{(\frac{x}{a})-1}}\left[\binom{n_{(\frac{x}{a})-1}}{n_{\frac{x}{a}}}0^{n_{\frac{x}{a}}}a^{n_{(\frac{x}{a})-1}-n_{\frac{x}{a}}}\right]\right]a^{c-d}\right]a^{b-c}\right]a^{r-b}\right]}^{\frac{x}{a}\text{ sums}}$$

you can stop at any layer you want, you dont need to fully expand

$$x^r = \overbrace{\sum_{b=0}^{r}\left[\binom{r}{b}\sum_{c=0}^{b}\left[\binom{b}{c}\sum_{d=0}^{c}\left[\binom{c}{d}\cdots\left[\sum_{n_l=0}^{n_{l-1}}\left[\binom{n_{l-1}}{n_l}(x-la)^{n_l}a^{n_{l-1}-n_l}\right]\right]a^{c-d}\right]a^{b-c}\right]a^{r-b}\right]}^{l\text{ sums}}$$

falling sums

$$(x+a)^r = \sum_{b=0}^{r}\left[\binom{r}{b}x^b a^{r-b}\right]$$

$$(x+a)^r = \sum_{b=0}^{r-1}\left[\binom{r}{b}x^b a^{r-b}\right] + \binom{r}{r}x^r a^0$$

$$((x-a)+a)^r = x^r = \sum_{b=0}^{r-1}\left[\binom{r}{b}(x-a)^b a^{r-b}\right] + \binom{r}{r}(x-a)^r a^0$$

the same reassoning applys here as in the last case:
"we can subtitute the defintion of x^r into $(x-a)^b$ or a^{r-b}
$(x-a)^b$ is a good substitution as it the nested power term will shrink by a
after $\frac{x}{a}$ expansions the inner power term will be 0 "

$$x^r = \sum_{b=0}^{r-1}\left[\binom{r}{b}(x-a)^b a^{r-b}\right] + \binom{r}{r}(x-a)^r a^0$$

$$(x-a)^b = \sum_{c=0}^{b-1}\left[\binom{r}{b}((x-a)-a)^c a^{b-c}\right] + \binom{b}{b}((x-a)-a)^b a^0$$

$$x^r = \sum_{b=0}^{r-1}\left[\binom{r}{b}\left[\sum_{c=0}^{b-1}\left[\binom{r}{b}((x-a)-a)^c a^{b-c}\right] + \binom{b}{b}((x-a)-a)^b a^0\right] a^{r-b}\right] + \binom{r}{r}(x-a)^r a^0$$

$$\overbrace{\sum_{b=0}^{r-1}\left[\binom{r}{b}\left[\sum_{c=0}^{b-1}\left[\binom{r}{b}\right.\right.\right.}^{l \text{ sums}}$$

$$\left.\left.\left.\ldots\left[\sum_{n_l=0}^{n_{l-1}-1}\left[\binom{n_{l-1}}{n_l}(x-la)^{n_l}a^{n_{l-1}-n_l}\right] + \binom{n_{l-1}}{n_{l-1}}(x-la)^{n_{l-1}}a^0\right]\ldots\right.\right.\right.$$

$$\left.\left.a^{b-c}\right] + \binom{b}{b}(x-2a)^b a^0\right] a^{r-b}\right] + \binom{r}{r}(x-a)^r a^0$$

$$x^r =$$

$$a^0 = 1 \text{ for all } a \in \mathbb{R}$$

$$\tbinom{r}{r} = 1 \text{ for all } r \in \mathbb{Z}^+$$

if we restrict our formula with these we can simplify it

according to our restriciton $a^0 = 1$ and $\binom{r}{r} = 1$

$$x^r = \overbrace{\sum_{b=0}^{r-1}\left[\binom{r}{b}\left[\sum_{c=0}^{b-1}\left[\binom{r}{b}\ldots\left[\sum_{n_l=0}^{n_{l-1}-1}\left[\binom{n_{l-1}}{n_l}(x-la)^{n_l}a^{n_{l-1}-n_l}\right] + (x-la)^{n_{l-1}}\right]\ldots a^{b-c}\right] + (x-2a)^b\right]a^{r-b}\right] + (x-a)^r}^{l \text{ sums}}$$

the sum can be factored into a sum of falling sums The papers theory holds if the restrictions hold:

$$a^0 = 1 \text{ for all } a \in \mathbb{R}$$

$\binom{r}{r} = 1$ for all $r \in \mathbb{Z}^+$ Since we can freely choose a only the restriction on r remains

“I have a theory, all x^n’s can be written as a sum of nested sums with falling upper bounds “

all x^r’s can be written as a sum of nested sums with falling upper bounds for all $r \in \mathbb{Z}^+$ also x must have a divisor d in $\mathbb{R}$ where $\left(\frac{x}{d} = l, l \in \mathbb{Z}^+\right)$, if it has this divisor it can be written a nested sum with l sums for all $a \in \mathbb{R} \mid r \in \mathbb{Z}^+ \mid d$ in $\mathbb{R}$ where $\left(\frac{x}{d} = l, l \in \mathbb{Z}^+\right) \Rightarrow$

$$x^r = \overbrace{\textstyle\sum_{b=0}^{r-1}\left[\binom{r}{b}\left[\sum_{c=0}^{b-1}\left[\binom{r}{b}\ldots\left[\sum_{n_l=0}^{n_{l-1}-1}\left[\binom{n_{l-1}}{n_l}(x-la)^{n_l}a^{n_{l-1}-n_l}\right]\right]\ldots a^{b-c}\right]\right]a^{r-b}\right]}^{l \text{ sums}}$$

$$+\overbrace{\textstyle\sum_{b=0}^{r-1}\left[\binom{r}{b}\left[\sum_{c=0}^{b-1}\left[\binom{r}{b}\ldots\left[\sum_{n_{l-1}=0}^{n_{l-2}-1}[(x-la)^{n_{l-1}}]\right]\ldots a^{b-c}\right]\right]a^{r-b}\right]}^{l-1 \text{ sums}}$$

$$+\ldots+\overbrace{\textstyle\sum_{b=0}^{r-1}\left[\binom{r}{b}[(x-2a)^b]a^{r-b}\right]}^{1 \text{ sums}}$$

$$+\overbrace{(x-a)^r}^{0 \text{ sums}}$$

Falling sums (x,r upper bound)

$$\sum_{b=0}^{r}\left[\binom{r}{b}x^b a^{r-b}\right](x+a)^r = \sum_{b=0}^{r-1}\left[\binom{r}{b}x^b a^{r-b}\right] + \binom{r}{r}x^r a^0 \text{ let } x' = x-a, \text{let } a' = a((x-a)+a)^r = \sum_{b=0}^{r-1}\left[\binom{r}{b}(x-a)^b a^{r-b}\right] + \binom{r}{r}$$

the previous falling sums resulted from expanding the inner (x-a)^b, if we expand the outer (x-a)^r we get a diffrent falling sum

$$((x-a)+a)^r = x^r = \sum_{b=0}^{r-1}\left[\binom{r}{b}(x-a)^b a^{r-b}\right] + \binom{r}{r}(x-a)^r a^0$$

$$x^r = \sum_{b=0}^{r-1}\left[\binom{r}{b}(x-a)^b a^{r-b}\right] + \binom{r}{r}\left[\sum_{b=0}^{r-1}\left[\binom{r}{b}((x-a)-a)^b a^{r-b}\right] + \binom{r}{r}((x-a)-a)^r a^0\right]a^0$$

$$x^r = \overbrace{\sum_{b=0}^{r-1}\left[\binom{r}{b}(x-a)^b a^{r-b}\right] + \binom{r}{r}\left[\sum_{b=0}^{r-1}\left[\binom{r}{b}((x-a)-a)^b a^{r-b}\right] + \binom{r}{r}\right.}^{l \text{ sums}}$$

$$\left.\cdots\left[\sum_{b=0}^{r-1}\left[\binom{r}{b}(x-la)^b a^{r-b}\right] + \binom{r}{r}(x-la)^r a^0\right]\cdots a^0\right]a^0$$

$$x^r = \overbrace{\sum_{b=0}^{r-1}\left[\binom{r}{b}(x-a)^b a^{r-b}\right]}^{l \text{ sums}}$$

$$+\binom{r}{r}a^0\sum_{b=0}^{r-1}\left[\binom{r}{b}(x-2a)^b a^{r-b}\right]$$

$$+\binom{r}{r}^2 a^{(0)^2}\sum_{b=0}^{r-1}\left[\binom{r}{b}(x-3a)^b a^{r-b}\right]$$

$$+\cdots+\binom{r}{r}^{l-1} a^{(0)^{l-1}}\sum_{b=0}^{r-1}\left[\binom{r}{b}(x-la)^b a^{r-b}\right]$$

$$x^r = +\binom{r}{r}^{l}(x-la)^r a^{(0)^l}$$

$$x^r = \sum_{b=0}^{l-1}\left[\binom{r}{r}^b a^{(0)^b}\sum_{c=0}^{r-1}\left[\binom{r}{c}(x-(b+1)a)^c a^{r-c}\right]\right] + \binom{r}{r}^{l}(x-la)^r a^{(0)^l}$$

note: for the simplifed sum aboe to work binom(r,r)^(0) and a^(0)^(0) must simplify to 1, so a^0 is implictly assumed to be 1 for that sum to work. you could factor out first term of the sum to make it pure

if we add restrictions the sum simplifies

$$a^0 = 1 \text{ for all } a \in \mathbb{R}$$

$$\binom{r}{r} = 1 \text{ for all } r \in \mathbb{Z}^+$$

$$\text{for all } a \in \mathbb{R} \mid r \in \mathbb{Z}^+ \mid l \in \mathbb{Z}^+$$

$$x^r = \sum_{b=0}^{l-1}\left[\sum_{c=0}^{r-1}\left[\binom{r}{c}(x-(b+1)a)^c a^{r-c}\right]\right] + (x-la)^r$$